Facility Sustainability Review: Library Energy and Daylighting Optimization

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FACILITY SUSTAINABILITY REVIEW: LIBRARY ENERGY AND DAYLIGHTING OPTIMIZATION

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Facility Code: BLD_LIB (Library Building)

Departmental Custody: Library Committee and Campus Facilities (DEP_ADMN).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. FACILITY PROFILE AND ARCHITECTURAL ATTRIBUTES

The Central Library building (BLD_LIB) encompasses 3,100 m2 of dedicated academic space spanning three levels. The building accommodates over 350 concurrent readers, featuring reading halls, periodical stacks, administrative reference desks, and an Advanced Digital Knowledge Portal equipped with 40 research terminals.

Architecturally, the building benefits from extensive floor-to-ceiling double-glazed fenestration along its eastern and northern facades, providing an abundance of natural daylight throughout morning and early afternoon hours. In 2024, the library consumed 168,200 kWh of electricity (54.2 kWh/m2/year), representing 7.12% of campus electrical consumption.

2. DAYLIGHT-HARVESTING SMART LIGHTING RETROFIT

In October 2023, the institution executed a deep lighting modernization across all three floors of BLD_LIB:

- LED Fixture Replacement: Removed 320 obsolete T8 fluorescent fixtures (36W + 10W magnetic ballast = 46W each) and installed 18W high-efficacy daylight-balanced LED luminaires (120 lumens/watt).

- Photocell Dimming Automation: Installed 12 photocell daylight sensors along the eastern perimeter reading zones. The smart lighting controller continuously measures ambient lux levels, automatically dimming artificial LED illumination when natural sunlight exceeds 400 lux.

- Energy Impact: The lighting retrofit reduced connected lighting load from 14.7 kW to 5.76 kW, saving 22,400 kWh annually and abating 18,368 kg CO2e based on CEA factor EF_ELEC_GRID (0.82 kg CO2e/kWh).

3. ROOFTOP SOLAR PV INTEGRATION (INIT_SOLAR_PH1)

Mounted on the terrace of BLD_LIB is a 30 kWp grid-tied monocrystalline solar photovoltaic array (75 modules of 400 Wp each) commissioned under Phase-1 Solar (INIT_SOLAR_PH1).

In 2024:

- Generated a verified 42,600 kWh of clean solar electricity.

- Directly satisfied 25.33% of the building's total annual electricity demand on-site.

- Avoided 34,932 kg CO2e of Scope 2 indirect emissions.

- Generated net electricity bill savings of INR 4,68,600.

4. DIGITALIZATION AND CIRCULAR RESOURCE CONSERVATION

Beyond energy optimization, BLD_LIB leads campus paper conservation:

- Institutional Digital Repository Portal: Transitioned 85% of scholarly journal subscriptions, conference proceedings, and past examination archives to digital cloud repositories accessed via terminals.

- Bound Print Reductions: Reduced institutional paper consumption for student project reports and syllabi by 65% since 2022, conserving an estimated 4.2 metric tonnes of paper annually.

- Avoided Embodied Emissions: Applying life-cycle paper manufacturing factors, paper digitization avoided 5,880 kg CO2e of embodied emissions.

These interventions establish BLD_LIB as a benchmark sustainable educational facility advancing SDG 7 (Affordable and Clean Energy), SDG 12 (Responsible Consumption and Production), and SDG 13 (Climate Action), supporting target TRG_NET_ZERO.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

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